

Human Motion Task



Fukun Zhang



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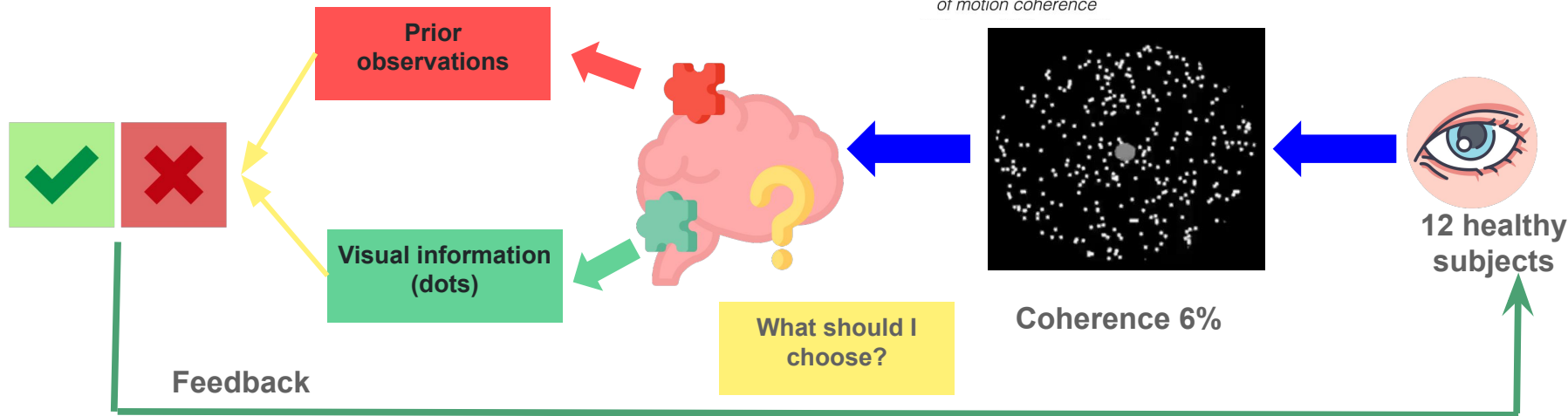
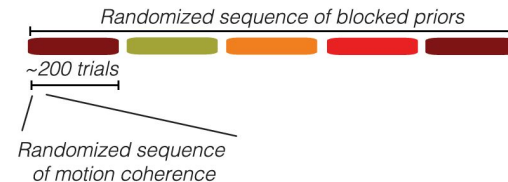
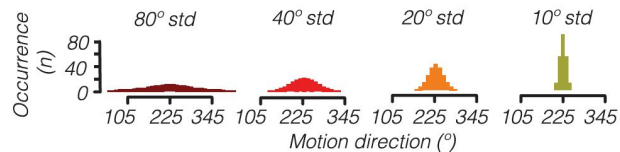


Janet Shin



Rosario Huaranca

Sensory Information Process

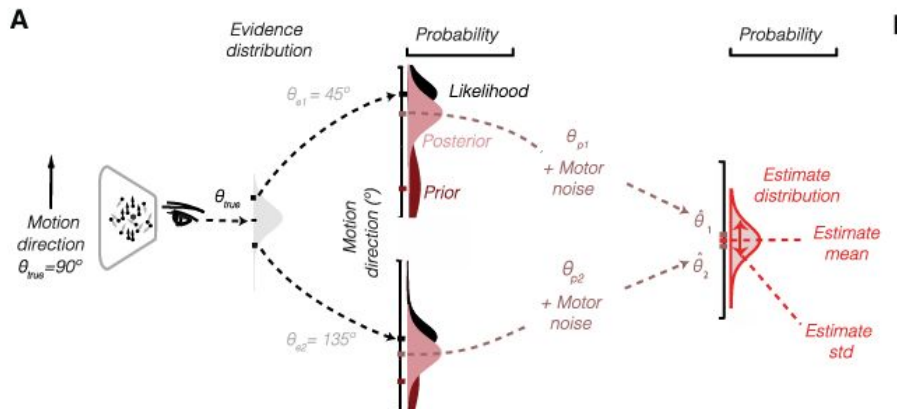


Objective: To test whether subjects' behavior conformed to the predictions of a Basic Bayesian observer.



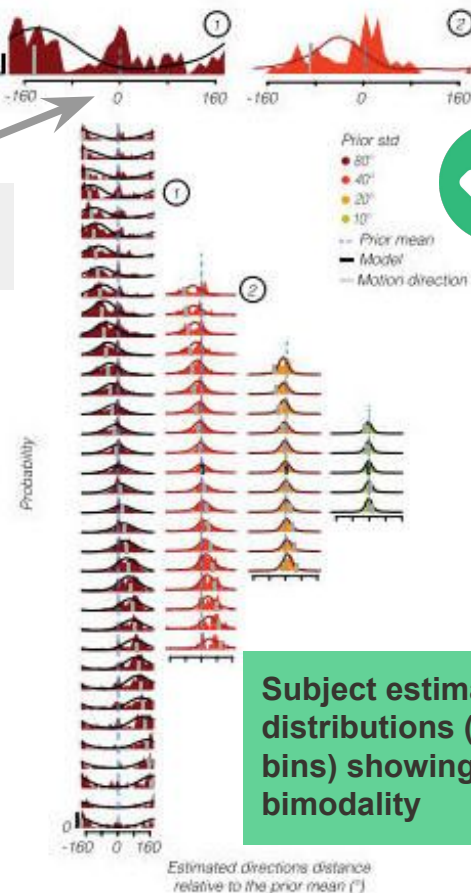
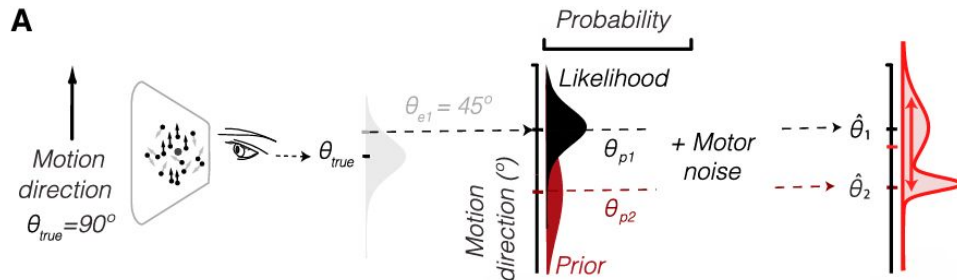
Basic Bayesian Model Observer

Why the classical model fails?



Result = Single unimodal peak??

The switching observer model



Subject estimate distributions (10° bins) showing bimodality

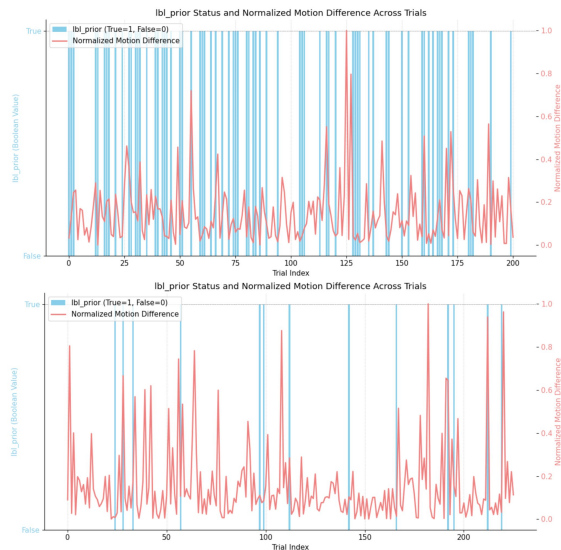


Our Research Question

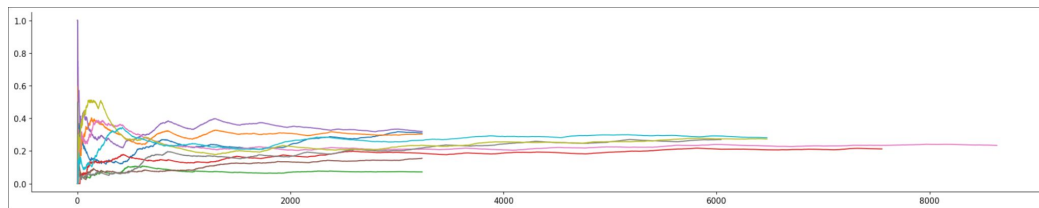
Assuming the switching model, what are specific factors/parameters that influence the decision for each subject between selecting prior or sensory (likelihood)?

Findings

Selection of priors with Errors



Cumulative Proportion of selecting priors



Key Findings:

Prior selection proportions remained stable post-learning, but varied significantly between subjects

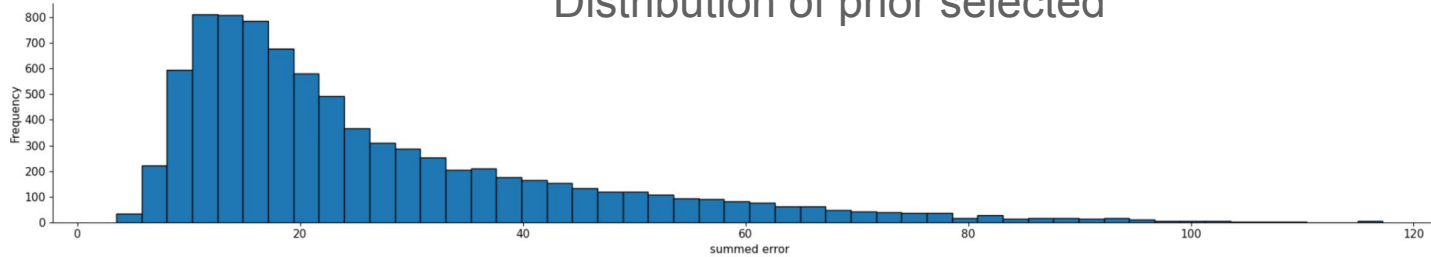
Key Findings:

Our attempt to correlate previous trial error with prior selection revealed substantial inter-subject variability in prior selection, making generalization difficult

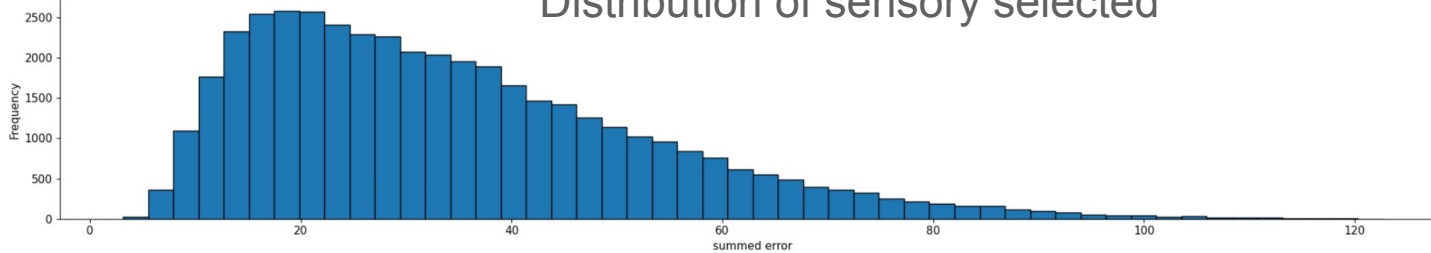


Distribution of the past 10 errors averaged

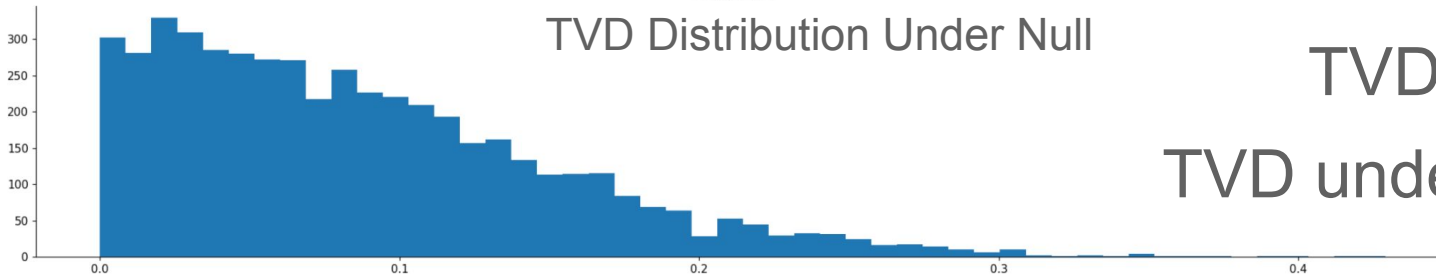
Distribution of prior selected



Distribution of sensory selected



TVD Distribution Under Null



TVD: 3.89

TVD under null: 0.05

Mixed-Effect Logistic Regression for Predicting Reliance on Sensory vs. Prior

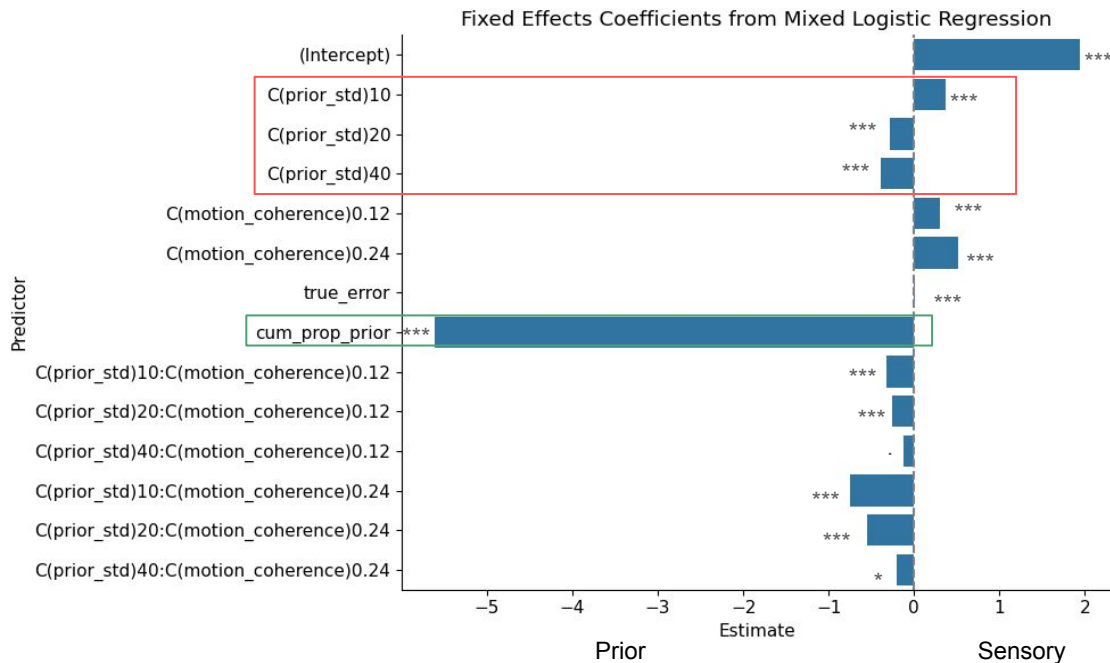
`chose_sensory~prior_std+coherence+prior_std*coherence+human_estimated_error
+cumulative_proportion_of_choosing_prior+(1|subject_id)`

- **chose_sensory**: binary outcome (1 = chose sensory; 0 = chose prior)
- **prior_std**: categorical levels of prior uncertainty
- **coherence**: categorical motion coherence level
- **human_estimated_error**: continuous predictor reflecting the distance between human motion and true sensory motion
- **cumulative_proportion_of_choosing_prior**: proportion of trials up to that point where the participant chose the prior
- **(1|subject_id)**: random intercept per subject

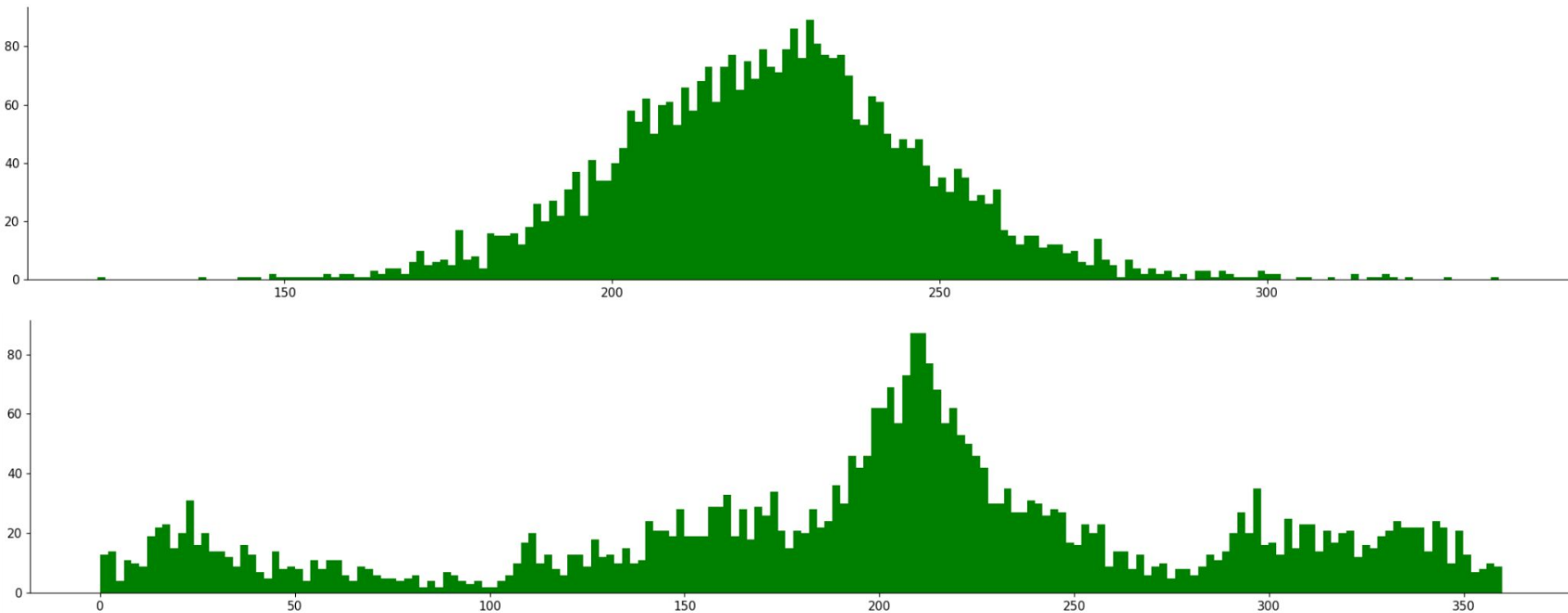
Reference group: prior_std = 80 and coherence = 0.06

Main takeaways:

- **Non-monotonic trend observed in prior_std**: comparing to prior_std=80, participants tend to use more sensory at prior_std=10, more prior at prior_std=20 & 40
- **Participants are less likely to switch to sensory when they have previously relied on prior**



Generative Model - Gaussian



Conclusions

- Counterintuitive results.
- Small sample size ($N=12$).
- Single Task scope.



Future directions

- Propose a **new generative model** in which a logistic regression, based on the most influential variables, computes the probability of relying on prior information versus sensory input. This probability guides a latent decision, upon which an estimation is drawn from one of two learnable distributions.

